Substitutes and Complements under Product Returns

Draft notes

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1 Model

A seller offers two goods separately and as a bundle. The price of good $i \in \{1, 2\}$ is p_i. The bundle price is p_B, with $\max(p_1, p_2) \leq p_B \leq p_1 + p_2$. Selling each good alone and selling the two together is mixed bundling. Producing good i costs c_i. Handling a return of good i costs the seller h_i.

There is a unit mass of consumers. Each consumer values good i at v_i. The values are drawn independently from the uniform distribution on $[0, 1]$ and are known before buying. Let W_i be the indicator that equals 1 when good i works and equals 0 otherwise. Good i works with probability $\alpha_i = \mathbb{E}[W_i]$. If good i works, the consumer receives v_i. If it does not work, the consumer receives zero. The values are independent of (W_1, W_2).

Let $\sigma_i = \sqrt{\alpha_i(1 - \alpha_i)}$ be the standard deviation of W_i, and let ρ be the correlation of W_1 and W_2. The probability that both goods work is $\alpha_1\alpha_2 + \rho\sigma_1\sigma_2$. The probability that only good 1 works is $\alpha_1(1 - \alpha_2) - \rho\sigma_1\sigma_2$. The probability that only good 2 works is $\alpha_2(1 - \alpha_1) - \rho\sigma_1\sigma_2$. The probability that neither works is $(1 - \alpha_1)(1 - \alpha_2) + \rho\sigma_1\sigma_2$. These four probabilities are positive. When $\rho = 0$ the goods work independently. A positive ρ means the goods are complements. They tend to work together. A negative ρ means the goods are substitutes. One good tends to work when the other does not.

The consumer buys good 1, good 2, the bundle, or nothing, before learning (W_1, W_2). Returning a good costs the consumer nothing. A consumer who keeps only good i pays p_i. A consumer who keeps both pays p_B. A consumer who keeps nothing pays nothing. The payoff from buying good i alone is $\max(W_i v_i - p_i,\, 0)$. The payoff from buying the bundle is

$$\max\big(W_1 v_1 + W_2 v_2 - p_B,\; W_1 v_1 - p_1,\; W_2 v_2 - p_2,\; 0\big).$$

The consumer chooses the option with the highest expected payoff. The consumer buys nothing when that payoff is negative.

The seller collects the price the consumer pays. The seller pays c_i for each good i the consumer buys and h_i for each good i the consumer returns. Profit is expected revenue minus these costs.

2 Consumer's decision

The expected payoff from buying nothing is zero. The expected payoff from buying good 1 is

$$\mathrm{EV}^1(v_1, v_2, p_1, p_2) = \alpha_1 \max(v_1 - p_1,\, 0).$$

The expected payoff from buying good 2 is

$$\mathrm{EV}^2(v_1, v_2, p_1, p_2) = \alpha_2 \max(v_2 - p_2,\, 0).$$

Suppose the consumer buys the bundle and only good i works. Keeping both pays $v_i - p_B$. This is no larger than $v_i - p_i$, because $p_B \geq p_i$. The good that does not work has value zero, and returning it pays nothing. Returning both goods pays zero. The payoff in this state is therefore $\max(v_i - p_i,\, 0)$. If neither good works, the payoff is zero. The expected payoff from the bundle is

$$\begin{aligned}\mathrm{EV}^b(v_1, v_2, p_1, p_2) = {} & (\alpha_1\alpha_2 + \rho\sigma_1\sigma_2)\max(v_1 + v_2 - p_B,\, v_1 - p_1,\, v_2 - p_2,\, 0) \\ & + (\alpha_1(1-\alpha_2) - \rho\sigma_1\sigma_2)\max(v_1 - p_1,\, 0) \\ & + (\alpha_2(1-\alpha_1) - \rho\sigma_1\sigma_2)\max(v_2 - p_2,\, 0).\end{aligned}$$

In every state, the bundle payoff is at least the payoff from having bought only one good. The consumer can keep that good and return the other when it works and is worth the price, and can return every good otherwise. When both goods work, keeping both can pay more. Thus $\mathrm{EV}^b(v_1, v_2, p_1, p_2)$ is at least as large as $\mathrm{EV}^1(v_1, v_2, p_1, p_2)$ and at least as large as $\mathrm{EV}^2(v_1, v_2, p_1, p_2)$. The consumer buys the bundle when $\mathrm{EV}^b(v_1, v_2, p_1, p_2)$ is positive. The consumer buys nothing when that value is zero.